

Appendix A. Proof of Claim 6

Fix $\tilde{\epsilon} > 0$ and note that for any $i \neq j \in [n]$, we have

$$\begin{aligned}
& \mathbb{E} \left[\frac{1}{(|\lambda_j - \lambda_i| + n^{-1.5-\tilde{\epsilon}})^2} \right] \\
&= \int_0^\infty \mathbb{P} \left(\frac{1}{(|\lambda_j - \lambda_i| + n^{-1.5-\tilde{\epsilon}})^2} > x \right) dx \\
&= \int_0^\infty \mathbb{P} \left(|\lambda_j - \lambda_i| < \frac{1}{\sqrt{x}} - n^{-1.5-\tilde{\epsilon}} \right) dx \\
&= \int_0^{n^{3+2\tilde{\epsilon}}} \mathbb{P} \left(|\lambda_j - \lambda_i| < \frac{1}{\sqrt{x}} - n^{-1.5-\tilde{\epsilon}} \right) dx + \int_{n^{3+2\tilde{\epsilon}}}^\infty \mathbb{P} \left(|\lambda_j - \lambda_i| < \frac{1}{\sqrt{x}} - n^{-1.5-\tilde{\epsilon}} \right) dx \\
&\leq \int_0^{n^{3+2\tilde{\epsilon}}} \mathbb{P} \left(|\lambda_j - \lambda_i| < \frac{1}{\sqrt{x}} \right) dx \leq 2c_0 n^{2.5+\tilde{\epsilon}},
\end{aligned}$$

where the last inequality holds by (Nguyen et al., 2017, Corollary 2.2) for some constant $c_0 > 0$. Using the above inequality, we get

$$\mathbb{E} \left[\sum_{i \neq j \in [n]: |i-j| \leq 2} \frac{1}{(|\lambda_j - \lambda_i| + n^{-1.5-\tilde{\epsilon}})^2} \right] \leq 8c_0 n^{3.5+\tilde{\epsilon}}.$$

Now, by the Markov's inequality, with probability $1 - n^{-\tilde{\epsilon}}$, we get

$$\sum_{i \neq j \in [n]: |i-j| \leq 2} \frac{1}{(|\lambda_j - \lambda_i| + n^{-1.5-\tilde{\epsilon}})^2} \leq 8c_0 n^{3.5+2\tilde{\epsilon}}.$$

As $\min_{i \in [n-1]} |\lambda_{i+1} - \lambda_i| \geq n^{-1.5-\tilde{\epsilon}}$ with probability $1 - o(1)$ by (Feng et al., 2019, Corollary 1), we get

$$\sum_{i \neq j \in [n]: |i-j| \leq 2} \frac{1}{(\lambda_j - \lambda_i)^2} \leq \sum_{i \neq j \in [n]: |i-j| \leq 2} \frac{4}{(|\lambda_j - \lambda_i| + n^{-1.5-\tilde{\epsilon}})^2} \leq 32c_0 n^{3.5+2\tilde{\epsilon}}. \quad (9)$$

Now, we focus on bounding the terms for which $|i - j| \geq 3$. By (Nguyen et al., 2017, Corollary 2.5) and union bound, we have $\min_{i \in [n-3]} |\lambda_{i+3} - \lambda_i| \leq n^{-6/5-\tilde{\epsilon}/5}$ with probability $1 - c_0 n^{-\tilde{\epsilon}}$ for some $c_0 > 0$. Using this bound, we get

$$\begin{aligned}
\sum_{i,j \in [n]: |i-j| \geq 3} \frac{1}{(\lambda_i - \lambda_j)^2} &\leq \frac{1}{\min_{i \in [n-3]} (\lambda_{i+3} - \lambda_i)^2} \sum_{i,j \in [n]: |i-j| \geq 3} \left[\frac{|j-i|}{3} \right]^{-2} \\
&\leq 6n^{17/5+2\tilde{\epsilon}/5} \sum_{i=1}^n \frac{1}{i^2} \leq \pi^2 n^{17/5+2\tilde{\epsilon}/5}.
\end{aligned}$$

Now, combining the above inequality with (9), we get with probability $1 - o(1)$

$$\sum_{i,j \in [n]: i \neq j} \frac{1}{(\lambda_j - \lambda_i)^2} \leq 32c_0 n^{3.5+2\tilde{\epsilon}} + \pi^2 n^{17/5+2\tilde{\epsilon}/5} \leq n^{3.5+3\tilde{\epsilon}},$$

where the last inequality holds for n large enough. Picking $\tilde{\epsilon} = \epsilon/6$ completes the proof.